

- (c) What are other examples of open sets that are simply connected?

22. Show that there is no holomorphic function f in the unit disc $\mathbb{D}$ that extends continuously to $\partial\mathbb{D}$ such that $f(z) = 1/z$ for $z \in \partial\mathbb{D}$.

9 Problems

1.* Consider a holomorphic map on the unit disc $f: \mathbb{D} \rightarrow \mathbb{C}$ which satisfies $f(0) = 0$. By the open mapping theorem, the image $f(\mathbb{D})$ contains a small disc centered at the origin. We then ask: does there exist $r > 0$ such that for *all* $f: \mathbb{D} \rightarrow \mathbb{C}$ with $f(0) = 0$, we have $D_r(0) \subset f(\mathbb{D})$?

- (a) Show that with no further restrictions on f , no such r exists. It suffices to find a sequence of functions $\{f_n\}$ holomorphic in $\mathbb{D}$ such that $1/n \notin f(\mathbb{D})$. Compute $f'_n(0)$, and discuss.
- (b) Assume in addition that f also satisfies $f'(0) = 1$. Show that despite this new assumption, there exists no $r > 0$ satisfying the desired condition.
[Hint: Try $f_\epsilon(z) = \epsilon(e^{z/\epsilon} - 1)$.]

The Koebe-Bieberbach theorem states that if in addition to $f(0) = 0$ and $f'(0) = 1$ we also assume that f is injective, then such an r exists and the best possible value is $r = 1/4$.

- (c) As a first step, show that if $h(z) = \frac{1}{z} + c_0 + c_1z + c_2z^2 + \cdots$ is analytic and injective for $0 < |z| < 1$, then $\sum_{n=1}^{\infty} n|c_n|^2 \leq 1$.

[Hint: Calculate the area of the complement of $h(D_\rho(0) - \{0\})$ where $0 < \rho < 1$, and let $\rho \rightarrow 1$.]

- (d) If $f(z) = z + a_2z^2 + \cdots$ satisfies the hypotheses of the theorem, show that there exists another function g satisfying the hypotheses of the theorem such that $g^2(z) = f(z^2)$.

[Hint: $f(z)/z$ is nowhere vanishing so there exists ψ such that $\psi^2(z) = f(z)/z$ and $\psi(0) = 1$. Check that $g(z) = z\psi(z^2)$ is injective.]

- (e) With the notation of the previous part, show that $|a_2| \leq 2$, and that equality holds if and only if

$$f(z) = \frac{z}{(1 - e^{i\theta}z)^2} \quad \text{for some } \theta \in \mathbb{R}.$$

[Hint: What is the power series expansion of $1/g(z)$? Use part (c).]

- (f) If $h(z) = \frac{1}{z} + c_0 + c_1z + c_2z^2 + \cdots$ is injective on $\mathbb{D}$ and avoids the values z_1 and z_2 , show that $|z_1 - z_2| \leq 4$.

[Hint: Look at the second coefficient in the power series expansion of $1/(h(z) - z_j)$.]

- (g) Complete the proof of the theorem. [Hint: If f avoids w , then $1/f$ avoids 0 and $1/w$.]

2. Let u be a harmonic function in the unit disc that is continuous on its closure. Deduce Poisson's integral formula

$$u(z_0) = \frac{1}{2\pi} \int_0^{2\pi} \frac{1 - |z_0|^2}{|e^{i\theta} - z_0|^2} u(e^{i\theta}) d\theta \quad \text{for } |z_0| < 1$$

from the special case $z_0 = 0$ (the mean value theorem). Show that if $z_0 = re^{i\varphi}$, then

$$\frac{1 - |z_0|^2}{|e^{i\theta} - z_0|^2} = \frac{1 - r^2}{1 - 2r \cos(\theta - \varphi) + r^2} = P_r(\theta - \varphi),$$

and we recover the expression for the Poisson kernel derived in the exercises of the previous chapter.

[Hint: Set $u_0(z) = u(T(z))$ where

$$T(z) = \frac{z_0 - z}{1 - \bar{z}_0 z}.$$

Prove that u_0 is harmonic. Then apply the mean value theorem to u_0 , and make a change of variables in the integral.]

3. If $f(z)$ is holomorphic in the deleted neighborhood $\{0 < |z - z_0| < r\}$ and has a pole of order k at z_0 , then we can write

$$f(z) = \frac{a_{-k}}{(z - z_0)^k} + \cdots + \frac{a_{-1}}{(z - z_0)} + g(z)$$

where g is holomorphic in the disc $\{|z - z_0| < r\}$. There is a generalization of this expansion that holds even if z_0 is an essential singularity. This is a special case of the **Laurent series expansion**, which is valid in an even more general setting.

Let f be holomorphic in a region containing the annulus $\{z : r_1 \leq |z - z_0| \leq r_2\}$ where $0 < r_1 < r_2$. Then,

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$$

where the series converges absolutely in the interior of the annulus. To prove this, it suffices to write

$$f(z) = \frac{1}{2\pi i} \int_{C_{r_2}} \frac{f(\zeta)}{\zeta - z} d\zeta - \frac{1}{2\pi i} \int_{C_{r_1}} \frac{f(\zeta)}{\zeta - z} d\zeta$$

when $r_1 < |z - z_0| < r_2$, and argue as in the proof of Theorem 4.4, Chapter 2. Here C_{r_1} and C_{r_2} are the circles bounding the annulus.

4.* Suppose Ω is a bounded region. Let L be a (two-way infinite) line that intersects Ω . Assume that $\Omega \cap L$ is an interval I . Choosing an orientation for L , we can define Ω_l and Ω_r to be the subregions of Ω lying strictly to the left or right of L , with $\Omega = \Omega_l \cup I \cup \Omega_r$ a disjoint union. If Ω_l and Ω_r are simply connected, then Ω is simply connected.

5.* Let

$$g(z) = \frac{1}{2\pi i} \int_{-M}^M \frac{h(x)}{x-z} dx$$

where h is continuous and supported in $[-M, M]$.

- (a) Prove that the function g is holomorphic in $\mathbb{C} - [-M, M]$, and vanishes at infinity, that is, $\lim_{|z| \rightarrow \infty} |g(z)| = 0$. Moreover, the “jump” of g across $[-M, M]$ is h , that is,

$$h(x) = \lim_{\epsilon \rightarrow 0, \epsilon > 0} g(x + i\epsilon) - g(x - i\epsilon).$$

[Hint: Express the difference $g(x + i\epsilon) - g(x - i\epsilon)$ in terms of a convolution of h with the Poisson kernel.]

- (b) If h satisfies a mild smoothness condition, for instance a Hölder condition with exponent α , that is, $|h(x) - h(y)| \leq C|x - y|^\alpha$ for some $C > 0$ and all $x, y \in [-M, M]$, then $g(x + i\epsilon)$ and $g(x - i\epsilon)$ converge uniformly to functions $g_+(x)$ and $g_-(x)$ as $\epsilon \rightarrow 0$. Then, g can be characterized as the unique holomorphic function that satisfies:

- (i) g is holomorphic outside $[-M, M]$,
- (ii) g vanishes at infinity,
- (iii) $g(x + i\epsilon)$ and $g(x - i\epsilon)$ converge uniformly as $\epsilon \rightarrow 0$ to functions $g_+(x)$ and $g_-(x)$ with

$$g_+(x) - g_-(x) = h(x).$$

[Hint: If G is another function satisfying these conditions, $g - G$ is entire.]

4 The Fourier Transform

Raymond Edward Alan Christopher Paley, Fellow of Trinity College, Cambridge, and International Research Fellow at the Massachusetts Institute of Technology and at Harvard University, was killed by an avalanche on April 7, 1933, while skiing in the vicinity of Banff, Alberta. Although only twenty-six years of age, he was already recognized as the ablest of the group of young English mathematicians who have been inspired by the genius of G. H. Hardy and J. E. Littlewood. In a group notable for its brilliant technique, no one had developed this technique to a higher degree than Paley. Nevertheless he should not be thought of primarily as a technician, for with this ability he combined creative power of the first order. As he himself was wont to say, technique without “rugger tactics” will not get one far, and these rugger tactics he practiced to a degree that was characteristic of his forthright and vigorous nature.

Possessed of an extraordinary capacity for making friends and for scientific collaboration, Paley believed that the inspiration of continual interchange of ideas stimulates each collaborator to accomplish more than he would alone. Only the exceptional man works well with a partner, but Paley had collaborated successfully with many, including Littlewood, Pólya, Zygmund, and Wiener.

N. Wiener, 1933

If f is a function on $\mathbb{R}$ that satisfies appropriate regularity and decay conditions, then its Fourier transform is defined by

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x \xi} dx, \quad \xi \in \mathbb{R}$$

and its counterpart, the Fourier inversion formula, holds

$$f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i x \xi} d\xi, \quad x \in \mathbb{R}.$$

The Fourier transform (including its d -dimensional variants), plays a ba-